



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.:7

Team No.: 17

Project Title: Kernel Monitor

Team Members

Roll Number	Student Name	Signature
2520030322	Challagulla Shravya	
2520030101	Gandu Rishitha	
2520030509	M.V.L.Samhita	

1. Abstract

The Kernel Monitor is a Linux-based C application developed to observe and display selected system and process information in real time. The current implementation collects overall CPU usage from /proc/stat, memory utilization and memory statistics from /proc/meminfo, and process IDs, names, and states by reading /proc and /proc/<PID>/status. It also monitors root filesystem storage using statvfs() and displays total, used, and free space. The monitor refreshes its terminal-based display every three seconds and uses predefined CPU and memory thresholds of 80% to generate alerts when limits are exceeded. Signal handling through SIGINT allows the monitor to be stopped safely using Ctrl+C. The project demonstrates practical Operating Systems concepts including process management, memory management, Linux /proc interfaces, file-system monitoring, signals, and system-level programming in C.

2. Problem Statement

Modern operating systems execute numerous processes and system operations simultaneously. Identifying resource-intensive processes, excessive CPU or memory usage, and abnormal system activity can be difficult using basic system information. Therefore, a Linux-based Kernel Monitor is required to continuously monitor important system and process-level activities and present them in an understandable format. The project focuses on applying Operating Systems and System Programming concepts to monitor system performance and resource utilization in real time.

3. Objectives

1. To monitor CPU, memory, and system resource utilization in real time.

2. To track running processes and their resource consumption.
3. To collect system-level information using Linux interfaces and system calls.
4. To identify excessive resource usage and provide useful alerts and monitoring information.

4. Proposed Methodology

The system will be developed on Linux using C/C++. It will collect system and process information from Linux kernel interfaces such as the /proc and /sys filesystems. The collected information will be processed to calculate CPU usage, memory utilization, process statistics, and other system parameters. The monitor will periodically refresh the collected data and display it through a simple interface. Threshold-based monitoring will be implemented to identify unusually high resource consumption. The overall flow will be data collection → processing → analysis → display → alert generation.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Detect process IDs, names, and states from /proc.
Memory Management	Calculate overall memory utilization and display RAM statistics.
/proc Filesystem	Obtain CPU, memory, process, and system information.
fopen(), fgets(), fscanf(), sscanf()	Read and parse information from Linux virtual files.
statvfs()	Obtain filesystem total, used, and free storage space.
Signals / SIGINT	Safely stop the monitor using Ctrl+C.
sleep()	Refresh the monitoring screen at a 3-second interval.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030322	Challagulla Shravya	Implement the process monitoring module by scanning /proc, detecting numeric PIDs, reading /proc/status, and displaying process IDs, names, and states for up to ten processes.
2520030101	Gandu Rishitha	Implement the CPU and memory monitoring module using /proc/stat and /proc/meminfo; calculate overall CPU and memory utilization and display total, free, and available memory.
2520030509	M.V.L.Samhita	Implement filesystem monitoring using statvfs(), threshold-based CPU/memory alerts, SIGINT signal handling, and the terminal refresh and monitoring flow.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment
C / C++	System-level programming
GCC Compiler	Compiling the source code
Linux /proc and /sys	Collecting system information
Git & GitHub	Version control and project repository
VS Code / Vim	Code development and editing

8. Expected Outcome

The project provides a working Linux-based terminal Kernel Monitor that periodically displays overall CPU usage, calculated memory usage, detected process IDs, process names and states, and root filesystem total, used, and free space. It applies predefined CPU and memory thresholds of 80% and generates alerts when these limits are exceeded. The monitor refreshes the display every three seconds and can be safely stopped using Ctrl+C. The project demonstrates practical Operating Systems concepts including process monitoring, memory monitoring, Linux /proc interfaces, filesystem monitoring, signals, and system-level programming in C.

9. GitHub Repository

GitHub Repository Link: <https://github.com/venkatalaxmisamhita/Kernal-Monitor>

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____